

B. Tech Degree III Semester Examination November 2011

SE 304 MANUFACTURING PROCESS (2006 Scheme)

Time : 3 Hours

Maximum Marks : 100

PART A

(Answer ALL questions)

(8 x 5 = 40)

- I. (a) What is meant by hardness? What are the different tests conducted to determine hardness?
(b) What is meant by creep? What are the important factors affecting creep with metals?
(c) Differentiate between TIG and MIG.
(d) What is called as weldability? What are the factors upon which the weldability depends?
(e) What are the different types of casting?
(f) What are the defects occurring during casting?
(g) Differentiate orthogonal and oblique cutting.
(h) What are the different lathe operations?

PART B

(4 x 15 = 60)

- II. Explain Iron-Carbon system by using Iron-Carbon Equilibrium diagram.

OR

- III. Write short notes on:

- (i) Annealing
- (ii) Normalising
- (iii) Hardening
- (iv) Tempering

- IV. Explain the principle of operation of submerged Arc welding and Gas metal arc welding.

OR

- V. List out the welding problems, their causes and remedies.

- VI. (a) Explain about moulds and core making in casting process.
(b) Explain the characteristics of cores.

OR

- VII. Explain about the Inspection and Testing of castings.

- VIII. Explain the mechanism of a Shaping machine. Describe about shaping machine operations.

OR

- IX. Write short notes on:

- (i) ECM
- (ii) EDM
- (iii) USM
- (iv) AJM

